

Mechanical Components & Mechanisms

If a component does any 1 of the 4: transfer energy, reduce energy loss, hold parts together, transmit/alter motion,
it is a mechanical component.

There are 6 overall groups of mechanical components

Fasteners & Locking Devices

Fasteners secure 2 or more parts together.

There are 2 main types of fasteners, Bolts and Nuts, and Screws

Bolts & Nuts

????????????????????????????????????

Screws

There are different properties of screws, being length, material, shape of head, thread type, diameter and slot type.

Screw Heads & Slot Type

these properties of the screw affect how the screw is installed.

the screw head type determines the tool used and also how the screw protrudes from the material

the slot type determines the tool used, and the wrong tool used may damage or strip the screw slot.



Hence, different screw types are named after the type of material is fastens together.

Wood Screws

designed for fastening wood

are self tapping, aka no threads have to be tapped or hole to be drilled (ONLY for softer woods) to insert the screw.

To make the screw self tapping, it has a sharp tip and a wide pitch (pitch is distance between screw threads), to ensure more material between threads

Made with zinc plated steel or stainless steel

Screws for Plastics

for fastening plastics

are self tapping, so no threads have to be tapped, however there has to be a pre-existing hole from drilling

hence, it has a flat tip instead, but still has a wide pitch.

Made with steel/stainless steel, coated with zinc or black oxide

Machine Screws

for fastening metal

not self tapping, so threads have to be tapped and holes have to be drilled before hand.

Hence, they have flat tips and narrow pitches.

Made with steel/stainless steel.

Bearings & Slides

There are 2 major types of bearings, each with 2 minor types as well

Linear Bearings

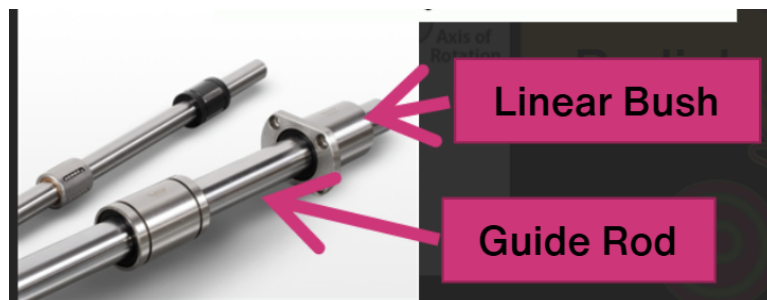
for reducing friction on linear sliding components

Circular Sliders

manages forces and moments in multiple directions,

allows parts to move along circular guide rods

economical but not very accurate

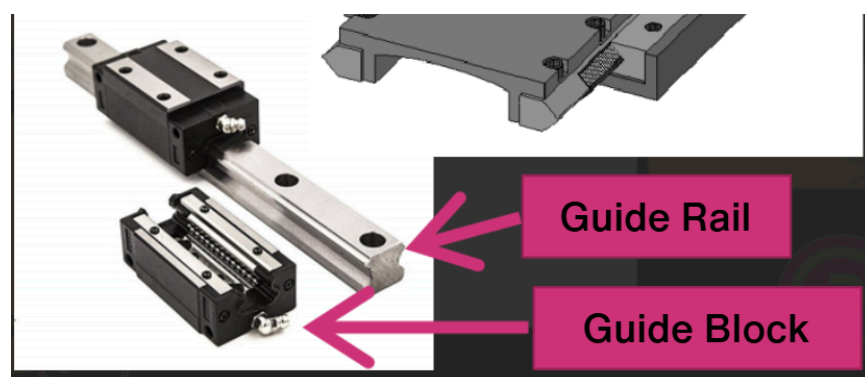


Linear Motion Guides

again, they manage forces and moments in multiple directions,

and allows part to move along guide rods

however, they are expensive but highly accurate and long lasting



Rotational Bearings

for reducing friction on rotating components

Axial Bearings

manages forces in the axial direction, e.g. in turn tables, where force is applied onto the axis downward.



hence they have independent surfaces on the top and bottom

Radial Bearings

manages forces in the radial direction (e.g. centrifugal forces, like in drills)



hence they have independent surfaces on the internal and external diameter

Springs

stores energy from moving components by working within its elastic deformation region

designed to operate in only 1 direction, and using them in other directions will damage, and plastically deform them

Compression

stores energy under compression, and release energy during expansion
used in train carriages, or in pens (for diff scales)

Tension

stores energy under tension, release energy during contraction
used in trampolines

Torsion

stores energy when twisted, releases energy when unravelled
used in clothes pegs, clips, or mousetraps

Flat Coil

a long tightly coiled strip of spring steel
stores energy when coiled OR uncoiled,
used in measuring tapes or watches.



Leaf Springs

multiple layers or pre bent metal strips stacked on top of one another
used to replace compression springs, when height is limited
hence used in automobiles.



Gears

transmits motion and power, by changing speed and torque.
As speed increases, torque decreases, and vice versa

in a gear system, the gear connected to a motor is the driver gear, while the output gear is the driven gear.

For multiple gears, the gears in between are also called idler gears, which may be used to have both driver and driven gears rotate in the same direction.

smth smth gear ratio affects torque and speed change wow !!!!!!!!!!!1

Spur Gears/Helical Spur Gears

for 2 parallel shafts.

straight teeth spur gears are economical, however not good for high power transmissions

while helical teeth reduce noise and vibration → hence used for high power transmission

Bevel Gears/Helical Bevel Gears

for 2 perpendicular shafts

like spur gears,

straight teeth are economical, however not good for high power transmissions

while helical teeth reduce noise and vibration → hence used for high power transmission

Worm & Worm Wheel

for 2 perpendicular shafts, where high gear ratio is needed (high speed → high torque)

is self locking, i.e. motion is transmitted only through worm to worm wheel, not the other way. making it useful for preventing unwanted movement when motor is not running

Rack & Pinion (straight/helical teeth)

converts rotational to linear motion, or vice versa.

can be in either straight or helical teeth (advantages said earlier)

Belts & Pulleys

transmits motion and power as well, similarly to gears.

Hence, as speed increases, torque decreases, and vice versa

and input pulley is driver pulley, output pulley is driven pulley.

However, pulleys do not change rotation direction.

There are different types of belts and pulleys:

Round Belt

- belt has a round cross section

- for 2 parallel or perpendicular shafts, and low torque applications

- pulley has raised sides to keep belt centered on pulley, preventing it from slipping off.

Flat Belt

- belt has a flat cross section

- used for 2 parallel shafts, with high speed & low torque applications

- larger belt surface area improves traction and prevents slipping, however pulley alignment is still important to prevent slippage

Vee Belt

- belt has a trapezium cross section

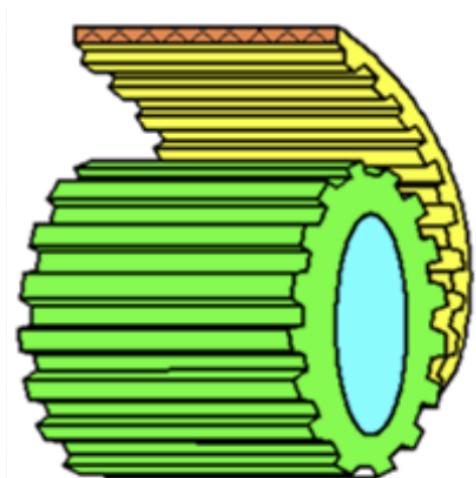
- used for 2 parallel shafts, with high speed & HIGH torque applications

- belt and pulley shape ensures belt stays in pulley, and large surface area helps with traction

- multiple belts may be used at the same time for reliability and safety

Timing Belt

- belt has notches that engage with notches in pulley.



for 2 parallel shafts, for high speed applications where accuracy is needed.

when the two notches engage with each other, slippage is eliminated, allowing for accurate transmission of motion (e.g. both pulleys engage at the same speed)

Conveyor Belt

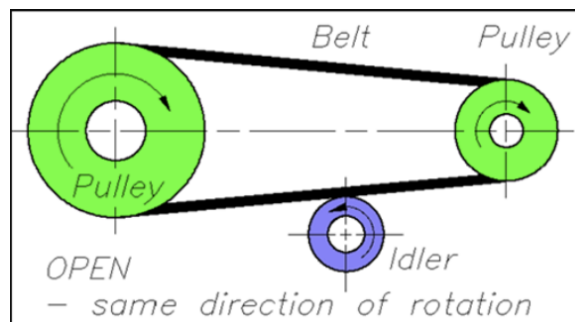
similar to flat belts,

for 2 parallel shafts for high speed applications to transport items.

With different pulleys and belts, they can be assembled differently.

Open

standard belt installation, input and output direction is the same, shafts are parallel



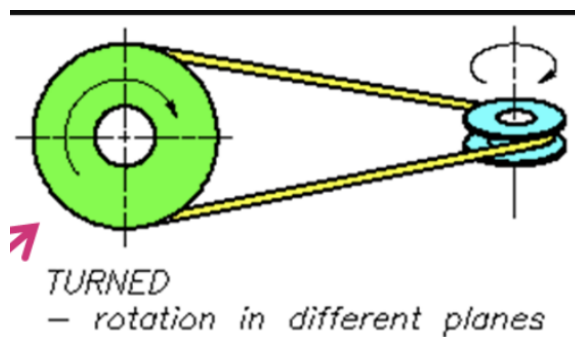
an idler pulley may be used to keep the belt taut.

for round, flat, vee and timing belts.

All other installations are only for round belts:

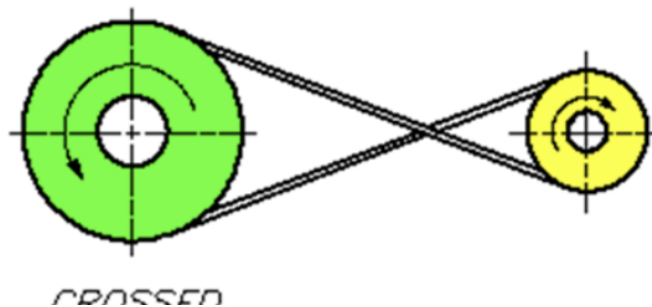
Turned

transmits motion to perpendicular shafts, keeping motion direction the same



Closed

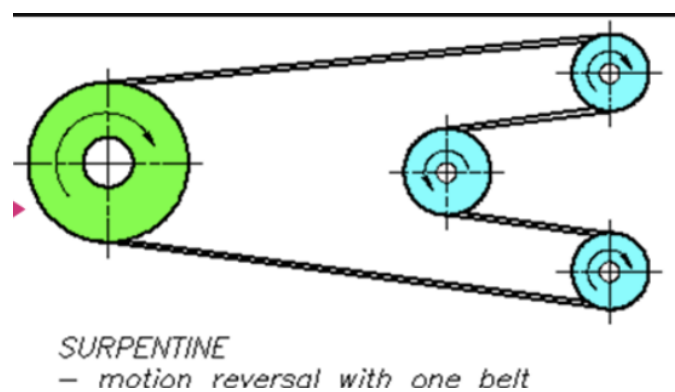
reverses motion direction on two parallel shafts



however crossing the belt causes rubbing which can damage the belt over time

Serpentine

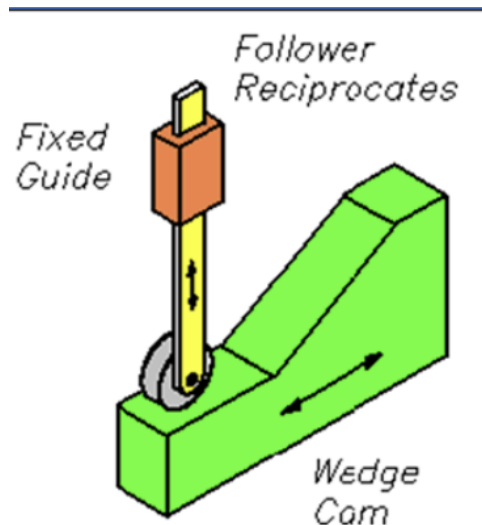
reverses motion direction on two parallel shafts w/o rubbing, however requires more pulleys



Cams & Followers

Cams and followers transmit a desired linear motion from a rotational motion (radial cams) or reciprocating linear motion (wedge cam)

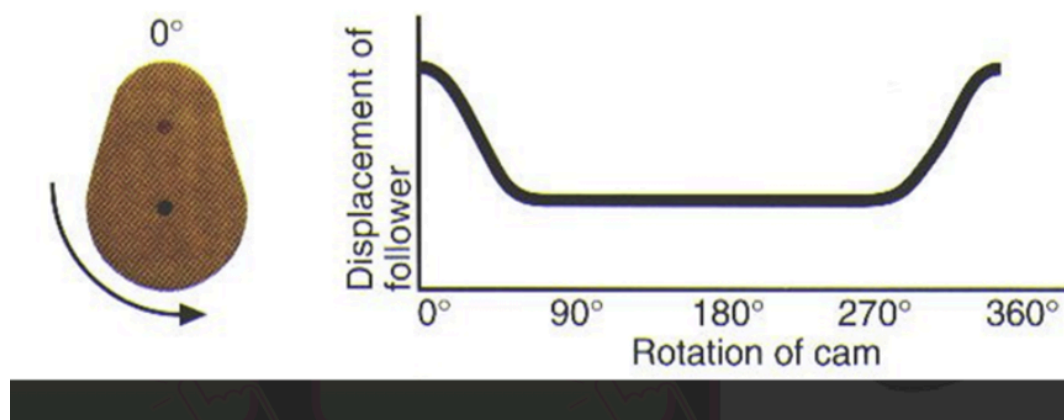
Wedge cams are the only type of cams that convert linear motion to another linear motion.



while there are 3 types of radial cams

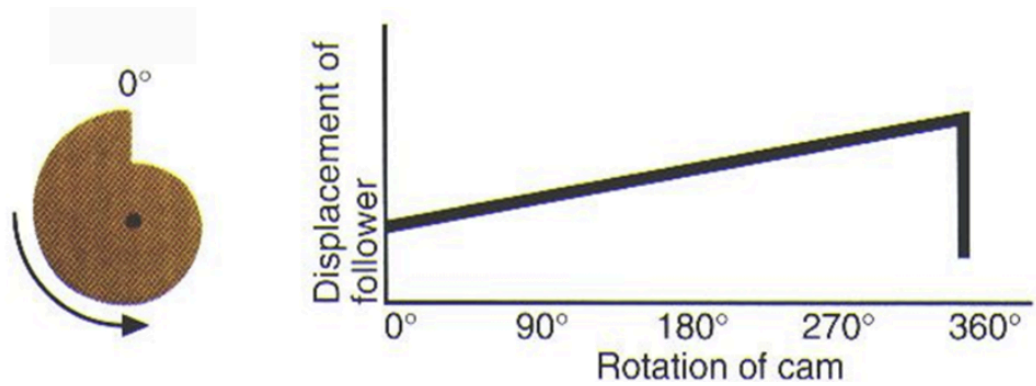
Pear

shape causes followers movement to gentle increase and decrease
large displacement for a small period, small displacement for a long period.



Snail

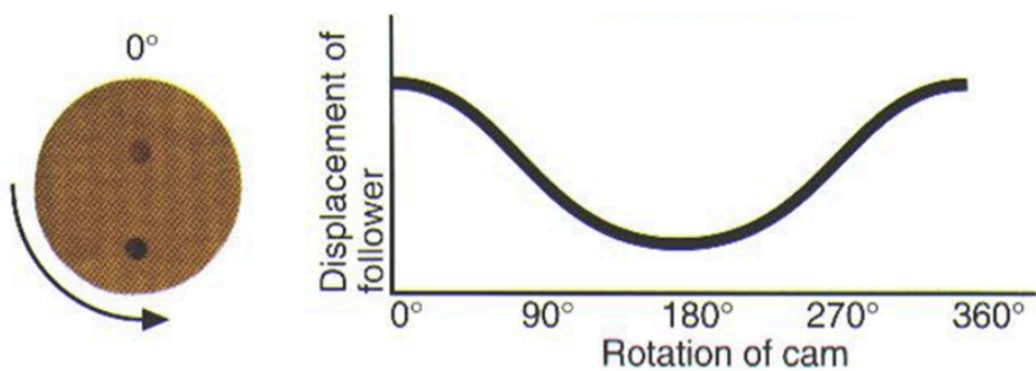
followers movement is a gentle increment before a sudden drop
displacement increases with time, until the sudden drop where it sharply changes.



Eccentric

follower moves in a sinusoidal wave.

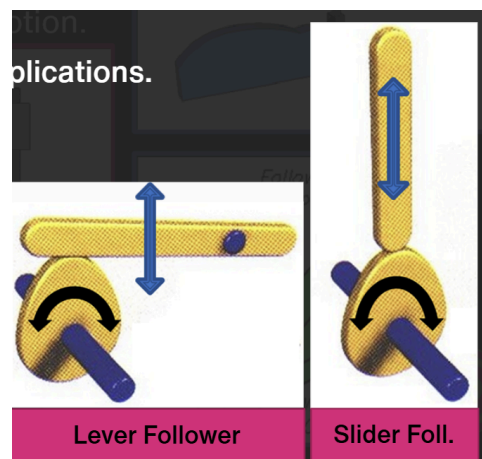
displacement varies across time



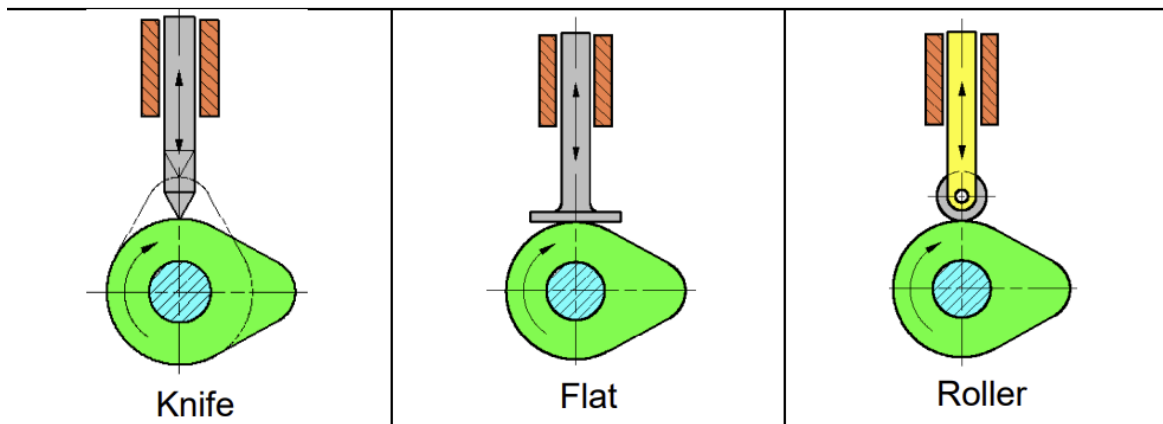
There are also different types of followers:

Slider followers, for when the motion needs to be directly above the cam, or

Lever followers, when the follower needs to be to the side of the cam.



There are also different types of follower ends, depending on the cam speed and cam shape complexity



Knife

for complex cam shapes and low speeds

Flat

for simple cam shapes and low speeds (and economical)

Roller

for simple cam shapes and fast speeds

Mechanisms are assemblies of mechanical components, usually giving a mechanical advantage.

Mechanical Advantage

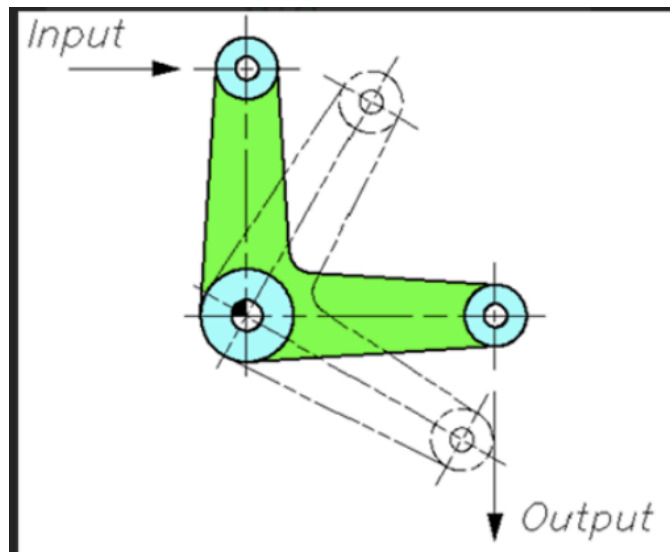
A ratio of output to input force, usually to obtain a greater output force, to make work easier and more efficient

Cranks

A flat piece connected to the end of a shaft, converting a large circular movement on the crank to rotation on the shaft, used in hand cranks or bicycle pedals

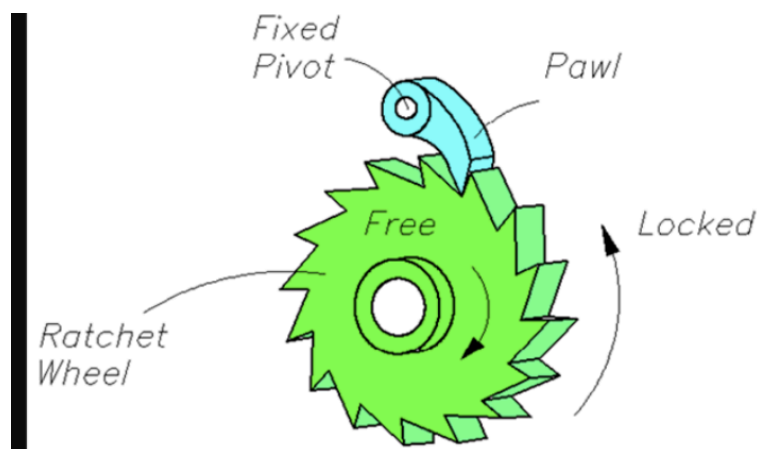
Bell Crank

a type of crank that transmits and redirects motion and force, like so



It also can have its arms changed to obtain a mechanical advantage
used race car suspensions, redirecting force to the suspension

Ratchet & Pawls



For one way rotation, where the pawl allows rotation of the ratchet in one direction, and blocks the ratchet teeth and prevents rotation in the opposite direction

used in safety winches, pulleys and rope, and bicycle wheel hubs (where the ratchet and pawl allows independent rotation of the wheel hub when not pedalling)

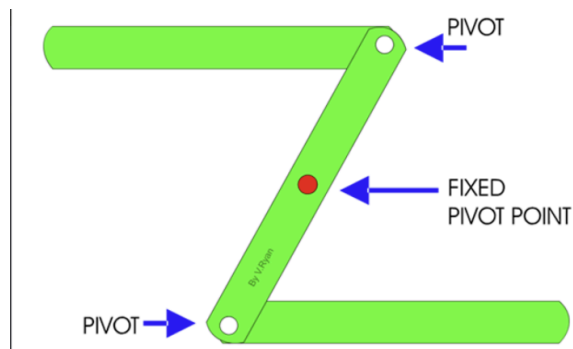
Linkages

a system of bars linked/fixed at points that they can rotate freely about, causing them to rotate simultaneously with each other

used for increasing mechanical advantage, by altering the output force and motion

there are different types of basic linkages:

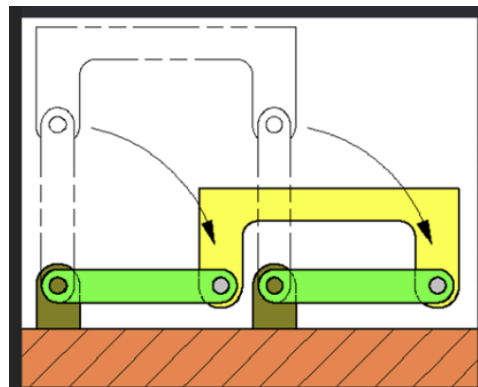
Reverse Linkage



reverses motion, keeps same force

useful for using one actuator/motor to move 2 objects in opposite directions

Parallel Linkage



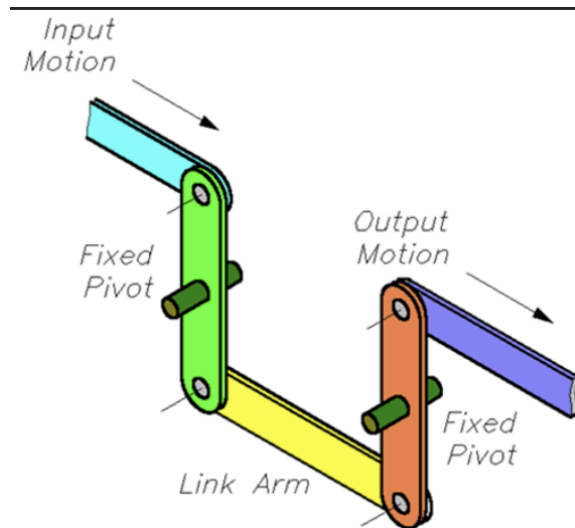
keeps one linkage parallel and at a fixed distance away from the fixed joint

useful for toolbox components (where compartments need to stay upright), or in width adjusting mechanisms

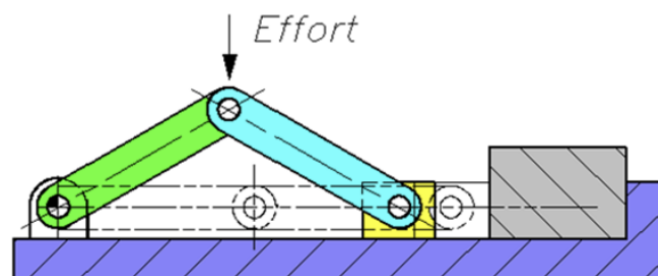
Push Pull Linkages

creates same input and output motion and force, useful when motion needs to be diverted,

used in rc helicopters

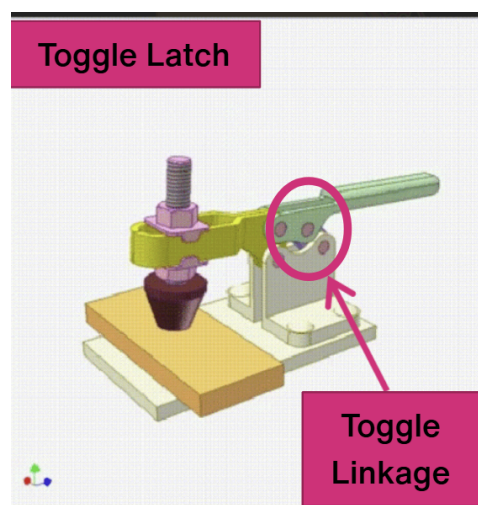


Toggle Linkages

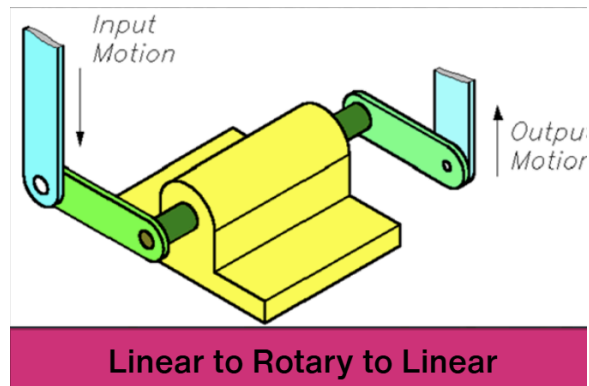


overcome large resistance (i.e. high force output) by applying a small input force over a long distance

used in toggle latches or tool box latches



Rotary Linkage



converts linear to rotational motion, or vice versa.

used when a motor needs to move an object linearly